# CHE 321 – Chemical Reaction Engineering
# Fall 2019

## Home Work – 6 (40 points)

## Due: 11:59 AM, 2nd Dec 2019

## Problem 1: The Shrinking Core Model

## Instructions

1) Email the explanation and figures in a single pdf file, and MATLAB codes as .m files to Lan.
2) Work independently while developing the MATLAB code.
3) Make necessary assumptions.
4) 5 points will be deducted for submissions after the deadline.

### Objectives

1) To understand the concept of shrinking core model.
2) To learn the technique of solving ODEs, PDEs using finite difference method in MATLAB.

- A spherical particle of radius Ro is participating in a single solid-fluid reaction. [$\propto S + A \rightarrow P$]
- The particle has initial radius Ro and is assumed to have retreated to radius R(t) during the course of reaction.
- Given:
  - Ro = 1 cm
  - De = $10^{-5}$ $cm^2/s$
  - $k_c$ = 0.001 m/s
  - $C_{Af}$ = 0.1 mol/L
  - $\hat{k}$ = $10^{-3}$ L/mol.s
  - $C_{So}$ = 20 mol/L
  - $\propto$ = $10^{-3}$ L/mol
- Obtain C (r, R(t)) and R(t) and plot these 2 graphs.

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## Problem 2:

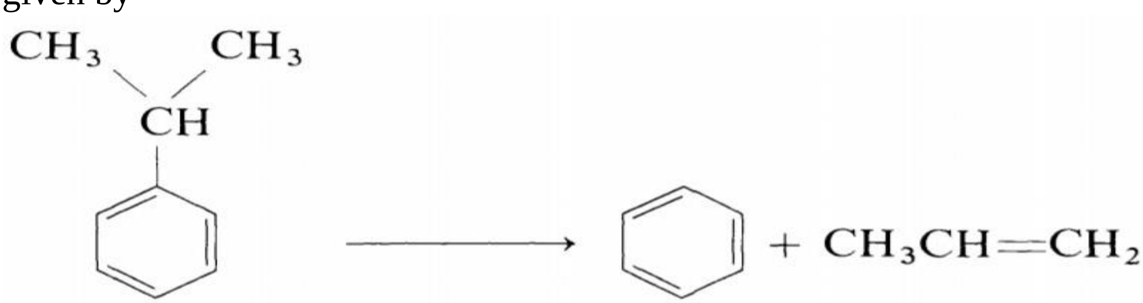
## Instructions

5) Email the explanation and figures in a single pdf file, and MATLAB codes as .m files to Lan.
6) Work independently while developing the MATLAB code.
7) Make necessary assumptions.
8) 5 points will be deducted for submissions after the deadline.

### Objectives

3) To understand the concept of effectiveness factor and to model and simulate a transient packed bed reactor.
4) To learn the technique of solving PDEs using finite difference method in MATLAB.

Consider the cracking reaction of Cumene into Benzene and Propylene which is given by



The reaction is exothermic ($\Delta H$ = -100000 J/mol) and it occurs in the presence of silica-alumina catalyst in a packed bed reactor of diameter 5 cm and length 50 cm. The specific volume remains constant. The surface area per unit volume of the catalyst pellet is 2000 $m^2/m^3$. Assume the reaction follows first order kinetics which is given by,

$r' = k' C_A$

where r' is the rate of the reaction per unit surface area ($mol/m^2.s$), k' denotes the rate constant (m/s) given by Arrhenius expression with a preexponential factor of 1 x $10^9$ m/s and an activation energy of 96450 J/mol. $C_A$ denotes the concentration of cumene at the catalyst surface ($mol/m^3$). All the pores are assumed to be uniform and they remain cylindrical of constant radius $r_o$ and length L. The reaction occurring on the catalyst is limited by the diffusional mass transport into the pore. The effectiveness factor for this system is 0.7. The reactant enters the reactor at a velocity of 1 m/s. The reactor is cooled with a cooling jacket maintained at 300 K and the overall heat transfer co-efficient is given as 100000 J/m$^2$ s K.

Consider a 1D transient model for the packed bed reactor. The mole balance is given by

$$\frac{\partial C_j}{\partial t} + v\,\frac{\partial C_j}{\partial z} = \sum_{i=1}^{m} \alpha_{ij} r_i + D\frac{\partial^2 C_j}{\partial z^2}$$

and the energy balance is given by

$$\rho C_p \frac{\partial T}{\partial t} + v\rho C_p \frac{\partial T}{\partial z} = K\frac{\partial^2 T}{\partial z^2} + \sum_{i=1}^{m} (-\Delta \mathrm{H}) r_i - \frac{4U}{d_t}\,(T - T_c)$$

Here $C_j$ denotes the concentration of species j, v denotes the velocity, $\rho$ = 2500 kg/m$^3$, $C_p$ = 680.73 J/kg.K, ΔH denotes the heat of reaction over the operating temperature range, U denotes the overall effectiveness factor, D denotes the diffusivity which is 1 x 10$^{-9}$ m$^2$/s, K denotes the thermal conductivity which is 1 W/m.K, $d_t$ denotes the tube diameter, $T_c$ denotes the temperature of the cooling fluid and $r_i$ denotes the rate of the reaction i (mol/m$^3$.s). Cumene enters the reactor at 490 K with an initial concentration of 0.1 mol/m$^3$. For the simulation, just use the mole balance for cumene and the energy balance.

1) Solve the steady state equations for the packed bed reactor by neglecting the axial diffusion using ode15s solver of MATLAB. Plot the concentration of cumene along the length of the reactor and the temperature along the length of the reactor. **(5 points)**

2) Solve the transient equations for the packed bed reactor by discretizing the convection term using a I order backward difference approximation and diffusion term using II order central difference approximation. Solve the resulting set of odes using ode15s solver of MATLAB. Plot the concentration of cumene and temperature along the length of the reactor at various times. Choose the total operating time to be 100s. **(10 points)**

3) Compare the steady state case in which the axial diffusion is neglected with the transient simulation case by plotting the concentration of cumene as a function of length obtained from the above cases in a single figure. Compare the temperature profiles also. For comparison with the steady state case, use the transient solution at infinite time. (**5 points**)